

J-ART: A Japanese Adversarial Red-Team Framework for Application-Layer LLM Security and Cost-Efficiency

Technical Report (v0.1, preprint)

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Abstract

Most open red-teaming and jailbreak benchmarks for large language models (LLMs) are English-centric and evaluate *bare models*. We present **J-ART** (Japanese Adversarial Red-Team framework), an open, reproducible harness that (i) evaluates the **application-structure layer** — model $\times$ system-prompt strength $\times$ guardrail $\times$ retrieval (RAG) — rather than bare models; (ii) applies **Japanese-specific obfuscation transforms** (gyaru-moji character substitution, vertical/newline splitting, keigo “polite-coercion” framing, double-talk false premises, Base64, and leet/zero-width smuggling) that defeat naïve keyword filters; (iii) grounds attacks in **MITRE ATLAS** (10 inference-time techniques); and (iv) reports a **cost-efficiency** metric alongside a defense rate with **Wilson 95% confidence intervals**. Attacks use *harmless proxy markers* (a fictitious canary and benign output markers); the framework contains **no weaponizable content**, and the malicious “core” of each attack is masked in all published artifacts. Across 21 application configurations and 11 attacks $\times$ 7 transforms (live, $K = 5$ trials per cell), we find that **application-layer defenses dominate raw model capability** — naked models range 29.6–89.6% defense, but a balanced ablation holding the model fixed shows **every guardrail and a hardened prompt significantly raise defense** (82% $\rightarrow$ 91–100%, all Newcombe CIs excluding zero) — and that a simple **normalizing filter** defeats the obfuscation transforms a naïve keyword filter misses. We also document **large run-to-run variance**: in a controlled $K = 5$ repeat of a frozen configuration set, **9.6% of cells changed outcome across identical trials**, which we argue makes single-sample leaderboards unreliable and motivates our statistical treatment.

Status. This is a preliminary technical report / preprint describing a proof-of-concept framework and exploratory findings. It is *not* an official safety evaluation of any vendor’s model. Numbers are run-time snapshots subject to the limitations in Section 7. References are indicative; bibliographic details should be verified before formal submission.

1 Introduction

Public LLM red-teaming tooling (e.g., garak) and jailbreak benchmarks (e.g., HarmBench, JailbreakBench) are predominantly English. Multilingual safety work has shown that *translating* prompts into low-resource languages can bypass safety training, but **language-specific obfuscation** — exploiting a language’s own scripts, typography, and register — remains comparatively underexplored. Japanese in particular offers a rich obfuscation surface (multiple scripts, full-width/zero-width characters, vertical writing, and conventionalized polite registers).

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Two further gaps motivate J-ART. First, **bare-model versus deployed-system evaluation**: practitioners deploy a model *plus* a system prompt, *plus* guardrails, often *plus* retrieval; the security-relevant unit is the **application configuration**, not the model. Second, **security and cost jointly**: defenses cost tokens or latency, so we report a cost-efficiency score so that “cheapest sufficiently-safe configuration” is directly readable.

Our contributions are: (a) a Japanese obfuscation transform suite; (b) an application-layer, MITRE-ATLAS-grounded harness with a safe proxy-marker methodology; (c) a guardrail comparison including a normalizing filter and a model-based classifier; (d) a statistically-reported leaderboard (Wilson CIs, optional repeated trials) with a cost-efficiency metric; and (e) an empirical observation of large router-induced run-to-run variance.

2 Related Work

J-ART sits at the intersection of six lines of work, summarized below and positioned in Table 1.

Adversarial red-teaming tooling. garak [2] scans a *model* with a library of vulnerability probes. J-ART instead treats the deployed *application configuration* (model $\times$ system prompt $\times$ guardrail $\times$ RAG) as the unit under test.

Jailbreak and harmful-behavior benchmarks. AdvBench/GCG [11], Jailbroken [12], HarmBench [3], JailbreakBench [4], and in-the-wild prompts [13] standardize *which* harmful behaviors to elicit and *how* to score them, predominantly in English at the model layer. J-ART reuses the harness idea but replaces harmful elicitation with a safe proxy-marker objective (§3) so the suite can be released publicly.

Automated attack generation. PAIR [6] and TAP [7] optimize jailbreak prompts against a target. J-ART uses a *fixed, auditable* transform suite, trading attack strength for reproducibility and safe disclosure.

Multilingual and low-resource safety. Deng et al. [14] and Yong et al. [5] show alignment degrades under *translation* into low-resource languages, and Wang et al. [15] study cross-lingual safety generalization. These study *translation across* languages; J-ART studies *intra-language orthographic obfuscation* specific to Japanese (gyaru-moji, vertical writing, keigo framing, kana/kanji encoding), which persists even though Japanese is high-resource.

Prompt injection. Perez & Ribeiro [16] introduced goal-hijacking and prompt-leaking; Greshake et al. [8] formalized *indirect* injection; Liu et al. [17] benchmark LLM-integrated apps. J-ART evaluates these as a function of the *application configuration*.

Guardrails and input classifiers. Llama Guard [9], NeMo Guardrails [18], and the OpenAI Moderation API [19] provide model- or rule-based filtering. J-ART treats the guardrail as a *swappable configuration variable* and measures the marginal defense each adds, including a deterministic normalizing filter that defeats obfuscations a naive keyword filter misses (§4).

LLM-as-judge validation. Zheng et al. [20] and the survey of Gu et al. [21] document the biases of LLM judges. J-ART avoids an LLM judge for its *primary* outcome, using a deterministic canary/marker check (§3) so judge variance does not enter the headline metric.

Table 1: Positioning of J-ART relative to representative prior work.

Work	Language	Unit of eval.	Cost axis	Guard/RAG	Public
garak [2]	English	Model	No	No	Yes
HarmBench / JBB [3, 4]	English	Model	No	No	Yes
PAIR / TAP [6, 7]	English	Model	No	No	Partial
Multiling. [5, 14, 15]	Low-resource	Model	No	No	Yes
Llama Guard / NeMo [9, 18]	Eng.-centric	Classifier	No	n/a	Yes
J-ART	Japanese	App config	Yes	Yes	Yes

Japanese NLP resources and taxonomies. General Japanese benchmarks such as JGLUE [22] cover NLU, but to our knowledge no open, reproducible *adversarial-safety* benchmark targets Japanese-specific obfuscation — the gap J-ART addresses. We map attacks to MITRE ATLAS [1] and cross-reference the OWASP Top 10 for LLM Applications [10].

J-ART is thus distinguished by combining a *Japanese intra-language obfuscation* surface, an *application-configuration* unit of evaluation, a *cost-efficiency* axis, and a *safe proxy-marker* methodology in a single reproducible harness.

3 Methodology

Application-structure layer. Each evaluated **target** is a configuration: provider/model $\times$ prompt_strength $\in \{\text{naked (low), hardened (high)}\} \times$ guardrail $\in \{\text{none, keyword, regex, llm, llamaguard}\} \times$ rag $\in \{\text{true, false}\}$. A simplified RAG context (benign corporate documents, optionally poisoned for indirect-injection attacks) and a system prompt containing a secret *canary* are assembled per trial.

Japanese obfuscation transforms. Each base attack is rendered through one of seven transforms: **baseline** (none), **polite_business** (keigo authority framing), **vertical_newline** (one-character-per-line vertical writing), **gyaru** (gyaru-moji glyph substitution + zero-width spaces), **double_tongue** (false sandbox/authorization premise), **base64_wrap** (Base64-encoded instruction), and **leet_smuggle** (leet symbol substitution + zero-width spaces). Transforms preserve human legibility while breaking substring-matching filters.

We distinguish two classes (Table 2). *Japanese-specific* transforms exploit the Japanese writing system or register and have no direct English equivalent: **gyaru** (kana glyph substitution), **vertical_newline** (vertical writing), and **polite_business/double_tongue** (keigo authority and in-group framing). *Language-agnostic* transforms — **base64_wrap** and **leet_smuggle**, plus the **baseline** control — apply to any language and serve as a comparison baseline so the marginal effect of the Japanese-specific surface can be isolated, clarifying which findings are expected to generalize beyond Japanese.

Attack suite and MITRE ATLAS mapping. Eleven attacks cover ten inference-time ATLAS techniques: Direct/Indirect Prompt Injection (AML.T0051.000/.001), LLM Jailbreak (AML.T0054), LLM Data Leakage (AML.T0057), LLM Prompt Self-Replication (AML.T0061), Trusted-Output/Citations manipulation (AML.T0067.000), LLM Prompt Obfuscation (AML.T0068), System-Prompt discovery (AML.T0069.002), RAG Poisoning (AML.T0070), and False RAG Entry Injection (AML.T0071). Vectors are **user** (direct) or **rag** (indirect).

Table 2: Classification of obfuscation transforms.

Class	Transforms	Exploits
Japanese-specific	<code>gyaru</code> , <code>vertical_newline</code> , <code>polite_business</code> , <code>double_tongue</code>	Kana glyph variants, vertical writing, keigo authority/in-group framing
Language-agnostic	<code>base64_wrap</code> , <code>leet_smuggle</code> <code>baseline</code>)	Encoding / symbol substitution applicable to any language (+

Harmless proxy markers and safety-by-construction. To avoid producing weaponizable content, attacks target **harmless proxies**: a fictitious canary passphrase embedded in the system prompt (data-leakage attacks) and benign string markers `[JART-PWNED]` / `[JART-HIJACKED]` (injection / control-hijack attacks). The malicious instruction **core** of each attack is defined in code, kept out of configuration files, and **masked** (replaced by a redaction placeholder) in `results.json` and the public site; only the obfuscation *framework* is shown, so developers can study the structure without a usable recipe. A defense-in-depth client-side mask is also applied at render time.

Deterministic judge. A breach is declared by exact membership: the canary in the response (`secret_leak`) or the marker in the response (`marker_injection` / `marker_hijack`). This judge is reproducible and cheap but conservative (it can miss semantically-equivalent leaks); see Section 7.

Guardrails. Five settings are compared: **none**; **keyword** (banned-substring match — fast, weak to obfuscation); **regex** (a *normalizing* filter that strips zero-width characters, reverses leet substitutions, folds whitespace, and Base64-decodes long tokens *before* keyword matching — deterministic, zero extra API cost); **llm** (the target model self-moderates input and screens output); and **llamaguard** (a dedicated safety-classification model — Llama Guard 4 via a model router — classifies the input as attack/benign, with a graceful fallback on failure).

Statistical treatment. Each model’s defense rate is reported with a **Wilson 95% confidence interval** computed over its trial count. Optionally, each cell (target, attack, transform) is repeated N times (`JART_TRIALS`, default 1); details aggregate per cell while summary statistics and CIs use the trial-level sample. In simulation mode each repeated trial is an independent Bernoulli draw, which narrows the CI as N grows (e.g., interval width $15.1 \rightarrow 7.5$ points from $N=1$ to $N=3$ for one configuration).

Cost-efficiency metric. We report $\text{cospa} = \text{defense_rate}(\%) \div \text{cost_per_million_tokens (USD)}$ (the per-million cost is clamped at a \$0.01 floor to avoid divergence), surfacing configurations that are both safe and cheap.

Execution. Trials are independent and run with bounded concurrency (a thread pool, default 8 workers) under per-request timeouts and exponential-backoff retries; determinism of the simulation does not depend on execution order. Targets whose API key is absent fall back to a deterministic simulation so a partial key set never aborts the run.

Table 3: Naked-model defense (no prompt, no guardrail; $K = 5$).

Naked model	Defense	95% CI
DeepSeek-V3	29.6%	[25.3, 34.4]
Llama 4 Scout	37.9%	[33.2, 42.9]
Gemini 2.5 Flash	49.6%	[44.6, 54.6]
gpt-oss-120b	82.6%	[78.5, 86.1]
GPT-4o-mini	83.9%	[79.9, 87.2]
Claude Opus 4.x	89.6%	[83.0, 93.8]

4 Results

We report a single, fully-labeled **live campaign** (21 configurations $\times$ 11 attacks $\times$ 7 transforms $\times$ $K = 5$ independent trials = 8,085 trials), executed 2026-06-14, total API cost **\$3.74**. Every cell is labeled **mode=LIVE** with a price snapshot; cells whose call ultimately failed are labeled **api_error** and **excluded from all aggregates**. Three rate-limited configurations (**gemini-pro**, **qwen3-max**, **claude-opus-4-8**) returned partial data (619 of 8,085 trials excluded), leaving 7,466 **clean trials**. The balanced-ablation numbers below come from a companion same-model run (**gpt-4o-mini**, $K = 3$, 0 errors).

Overall. The aggregate breach rate over clean trials was **14.8%** (1,108/7,466) and was **concentrated in weak configurations** — low-capability naked models. Every hardened or guardrailed configuration, on *any* model, defended in the **91–100%** band, whereas bare models ranged from **29.6% to 89.6%**.

Application-layer defense vs. model capability. Naked defense depends heavily on the model — a **60-point spread** (Table 3). The **balanced ablation** holds the model fixed (**gpt-4o-mini**) and crosses {naked, hardened} prompt $\times$ guardrail, testing each cell’s difference from the naked baseline (82.2%, [76.8–86.6]) with a Newcombe 95% CI. *Every* added defense is significant (CI excludes 0): keyword **+9.1%** [+2.9,+15.3], regex **+10.0%** [+3.9,+16.1], Llama Guard **+15.6%** [+10.4,+21.2], and a hardened prompt or LLM guardrail **+17.7%** [+13.1,+23.2], the last two reaching **100%**. Combining the two: a hardened cheap model (**gpt-4o-mini** hardened = 100%) **out-defends the strongest naked model** (Opus naked = 89.6%) at a fraction of the cost. The thesis — *configuration matters more than model choice* — is thus supported both within a fixed model and across the naked spread, not by a confounded single comparison.

Run-to-run variance (a primary result). We replace the earlier confounded two-run comparison with a **controlled experiment**: a single frozen configuration set re-run $K = 5$ times, everything else held constant. **9.6% of clean cells** (142/1,477) **changed outcome across the five identical repeats**, concentrated in naked weak models (DeepSeek-V3: 37 unstable cells; Llama 4 Scout: 32). For “Llama 4 Scout / naked” — the original cautionary example — of its 77 cells, **32 (42%) were unstable**: 14 never breached (0/5), 31 always (5/5), and **32 in between** (e.g. seven at 2/5, three at 3/5, thirteen at 4/5). A *single-sample* leaderboard would assign this exact configuration anywhere from 0% to 100% defense by chance. This is controlled evidence — free of the configuration-set confound — that **single-sample LLM leaderboards are not reproducible**.

Guardrail comparison. From the fixed-model ablation (**gpt-4o-mini**), each input guardrail raises defense above the 82.2% naked baseline (Table 4): keyword 91.3%, regex 92.2%, Llama Guard 97.8%, and LLM / hardened-prompt 100%. Guardrails that block an input **skip the**

Table 4: Guardrail effect, model held fixed (**gpt-4o-mini** ablation).

Configuration	Defense	Δ vs. naked (95% CI)
naked (baseline)	82.2%	—
+ keyword	91.3%	+9.1% [+2.9, +15.3]
+ regex (normalizing)	92.2%	+10.0% [+3.9, +16.1]
+ Llama Guard	97.8%	+15.6% [+10.4, +21.2]
+ hardened / LLM	100.0%	+17.7% [+13.1, +23.2]

main model call, lowering net cost while raising defense. The **normalizing regex filter** deterministically neutralizes the vertical-newline, leet, Base64, and gyaru transforms the keyword filter misses — verified in unit tests — at zero additional API cost. On the cost-efficiency metric, leaders are inexpensive OSS models behind a lightweight guardrail (gpt-oss-20b + keyword, cospa $\approx$ 1131), while a flagship naked configuration is worst (Opus naked, cospa $\approx$ 7.4) — three orders of magnitude apart.

Attack and transform effectiveness. The most effective attacks were Trusted-Output/Citations manipulation (**30.2%** breach), False RAG Entry Injection (24.1%), and System-Prompt discovery (20.3%). Among transforms, the *un-obfuscated baseline* was most effective (**22.2%**), then *vertical_newline* (20.5%) and *polite_business* (17.7%), while **base64_wrap** was **least effective (5.9%)**: heavy encoding makes models refuse or fail to parse the instruction more often than it smuggles it past them. The most dangerous Japanese transforms are the *legible* ones (vertical writing, keigo framing), not the cryptographic ones.

5 The Leaderboard Artifact

The harness emits **results.json** and a self-contained static site (sortable leaderboard, per-configuration attack logs with masked prompts, JP/EN UI with environment-based auto-selection, per-model defense-rate CIs, and per-cell breach rates), continuously deployed.

6 Ethics and Responsible Disclosure

Safety by construction. J-ART is safe by construction. (1) Attacks target a **fictitious canary** and **benign markers**, never real harmful capabilities; no weaponizable content exists in the code, data, or published site. (2) The malicious instruction **core** is masked in all artifacts; only the obfuscation framework is shown. (3) Models are referenced for research comparison only; results are run-time snapshots, not certifications. The intended use is to help developers harden their *own* deployments.

A reusable disclosure pattern. We state the design above as a transferable pattern for publishing adversarial safety benchmarks without releasing operational attacks: (i) a *harmless proxy objective* (a fictitious canary and benign markers) so a breach is measurable but inert; (ii) a *core/framework split with masking* — publish the obfuscation framework that carries the research signal, mask the operational core in every artifact, and keep harmless cores only in code; (iii) a *deterministic offline mode* so results reproduce from the repository without distributing a working exploit corpus; and (iv) *audited masking* — a unit-test invariant checks that no core (raw or obfuscated) reaches a published artifact. This keeps the methodology and aggregate findings fully open while the attack payloads stay withheld — the posture we recommend for language-specific red-team datasets generally.

Table 5: Governance crosswalk (informational).

J-ART (ATLAS)	capability	OWASP LLM Top 10	NIST RMF	AI	EU AI Act	Japan
Prompt injection / jailbreak (T0051, T0054)		LLM01	MEASURE 2.7		Art. 15, 55	AISI; AI Biz
Canary / data leakage (T0057)		LLM02	MEASURE 2.7		Art. 15, 10	AI Biz
System-prompt discovery (T0069.002)		LLM07	MEASURE 2.7		Art. 15	AISI
RAG poisoning / false entry (T0070, T0071)		LLM08; LLM01	MEASURE 2.7		Art. 15, 10	AISI
Trusted-output manipulation (T0067.000)		LLM09; LLM05	MEASURE 2.6		Art. 50	AI Biz
Obfuscation / self-replication (T0068, T0061)		LLM01	MEASURE 2.7		Art. 55	AISI

Governance crosswalk. Table 5 cross-references each evaluated capability to MITRE ATLAS [1], the OWASP Top 10 for LLM Applications [10], the NIST AI RMF [23], the EU AI Act [24], and Japan’s AI governance guidance [25, 26]. The mapping is **informational, not a compliance claim**; applicability depends on each deployment’s risk classification. Operationally J-ART is a **MEASURE**-stage activity in the NIST AI RMF whose outputs inform an organization’s **MANAGE** decisions.

7 Limitations

- **Proxy markers, not harm:** we measure instruction-violation / leakage proxies rather than real-world harm.
- **Deterministic judge:** the judge matches the canary/marker exactly *and* after light normalization (case-folding, whitespace/zero-width removal), so a reformatted echo is counted as a breach. It cannot detect *semantic*/paraphrased leakage, so reported defense rates are an **upper bound**; a human-validated LLM-judge is future work.
- **Single-turn:** multi-turn / crescendo attacks are not yet modeled.
- **Sample size:** a handcrafted suite (11 attacks $\times$ 7 transforms); default $N=1$ per cell.
- **Provider non-determinism / router variance:** results are snapshots and depend on routing and model versions at run time.
- **Partial data for rate-limited configs:** in the live campaign three configurations (Gemini-pro, Qwen3-max, Opus) hit provider rate limits; their failed cells are labeled **api_error** and excluded, so their estimates rest on fewer trials ($n = 116\text{--}295$ vs. 385) with wider CIs.
- **Cost attribution:** guardrail tokens are now priced at the *guardrail model’s* own rate (the classifier JART_GUARD_MODEL for llamaguard; the target model itself for the llm guardrail, which genuinely calls it) and reported separately. Remaining approximation: simulated token *counts* are estimated.
- **Simulation mode** is a deterministic stand-in for missing keys and must not be read as empirical model behavior.

8 Future Work

LLM-judge with human-validated agreement; multi-turn and adaptive (PAIR/TAP-style) attacks; larger and partially human-curated attack sets; additional ATLAS techniques (cost-exhaustion, model extraction); per-provider concurrency and incremental caching with longitudinal trend tracking; and dropping simulation mode for a purely empirical, repeated-trial statistical report.

9 Reproducibility

Code, configuration, and the deployment workflow are public. The harness is deterministic in simulation mode; live results depend on provider routing and model versions at run time and should be reported with the run date and model identifiers. See the repository for exact model identifiers and pricing used. Each cell carries provenance labels — `mode` (LIVE/MOCK), `api_error`, the resolved model, and a `price_per_million` snapshot — and simulated cells are excluded from the live aggregates. The reviewer- response experiments for the balanced ablation and the controlled variance study are scripted under `experiments/` with their statistics (`wilson_diff_ci`) and a cost estimate; the **labeled live campaign** that fills in the definitive numbers is a single repeated, fully-labeled run of that scaffolding.

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